

Comprehensive Mathematics Reference: Logic, Complex Numbers, Sequences, and Functions

Compiled from Chapter Notes

Contents

1 Chapter 1: Logical Arguments and Formal Proofs / Chapitre 1 : Arguments logiques et preuves formelles	3
1.1 Basic Terminology / Terminologie de base	3
1.2 Proposition and Truth Values / Proposition et valeurs de vérité	3
1.3 Logical Operations / Opérations logiques	3
1.4 Basic Logical Laws / Lois logiques de base	4
1.5 Quantifiers / Quantificateurs	4
1.5.1 Negation of Quantified Propositions / Négation des propositions quantifiées	4
1.6 Proof Techniques / Techniques de preuve	5
1.7 Real Number System / Système des nombres réels	5
1.7.1 Algebraic Properties of $\mathbb{R}$ / Propriétés algébriques de $\mathbb{R}$	5
1.7.2 Order Structure of $\mathbb{R}$ / Structure d'ordre de $\mathbb{R}$	6
2 Chapter 2: Complex Numbers / Chapitre 2 : Nombres complexes	6
2.1 Definitions / Définitions	6
2.2 Algebraic Operations / Opérations algébriques	6
2.3 Conjugate / Conjugué	7
2.4 Modulus (Absolute Value) / Module (Valeur absolue)	7
2.5 Geometric Representation / Représentation géométrique	7
2.6 Polar Form / Forme polaire	8
2.7 De Moivre's Formula / Formule de De Moivre	8
2.8 n-th Roots / Racines n-ièmes	8
3 Chapter 3: Sequences of Real Numbers / Chapitre 3 : Suites de nombres réels	9
3.1 Definitions / Définitions	9
3.2 Convergence / Convergence	9
3.3 Limit Theorems / Théorèmes sur les limites	9
3.4 Monotone Sequences / Suites monotones	10
3.5 Series / Séries	10
3.6 Subsequences / Sous-suites	10

3.7	Adjacent Sequences / Suites adjacentes	10
3.8	Cauchy Sequences / Suites de Cauchy	10
3.9	Recurrence Sequences / Suites récurrentes	11
4	Chapter 4: Functions / Chapitre 4 : Fonctions	12
4.1	Basic Notions / Notions de base	12
4.1.1	Bounded Functions / Fonctions bornées	12
4.1.2	Monotonicity / Monotonie	12
4.1.3	Composition / Composition	12
4.1.4	Special Functions / Fonctions spéciales	13
4.2	Limits of Functions / Limites de fonctions	13
4.2.1	Sequential Criterion / Critère séquentiel	13
4.2.2	Indeterminate Forms / Formes indéterminées	13
4.3	Continuity / Continuité	13
4.3.1	Discontinuities / Discontinuités	14
4.3.2	Sequential Continuity / Continuité séquentielle	14
4.3.3	Extension by Continuity / Prolongement par continuité	14
4.3.4	Intermediate Value Theorem (IVT) / Théorème des valeurs intermédiaires	14
4.3.5	Uniform Continuity / Continuité uniforme	14
4.4	Differentiability / Dérivabilité	14
4.4.1	Derivative Rules / Règles de dérivation	15
4.4.2	Higher-order Derivatives / Dérivées d'ordre supérieur	15
4.4.3	Mean Value Theorems / Théorèmes de la moyenne	15
4.4.4	L'Hôpital's Rule / Règle de L'Hôpital	15
4.4.5	Taylor-Lagrange Theorem / Théorème de Taylor-Lagrange	15

1 Chapter 1: Logical Arguments and Formal Proofs / Chapitre 1 : Arguments logiques et preuves formelles

1.1 Basic Terminology / Terminologie de base

- **Axiom / Axiome:** A statement that is given to be true.
- **Rule of Inference / Règle d'inférence:** A logical rule used to deduce one statement from others.
- **Theorem / Théorème:** A proposition that can be proved using definitions, axioms, other theorems, and rules of inference.
- **Lemma / Lemme:** A "pre-theorem" or a result needed to prove a theorem.
- **Corollary / Corollaire:** A "post-theorem" or a result that follows from a theorem, lemma, or another corollary.

1.2 Proposition and Truth Values / Proposition et valeurs de vérité

Definition 1. A **proposition** is a mathematical statement that is either true or false, but not both.

- **Truth Value / Valeur de vérité:** The truth or falsity of a proposition. Denoted as true (T) or false (F).
- **Truth Table / Table de vérité:** A table showing the truth value of a proposition based on the truth values of its components.
- **Equivalence / Équivalence:** Two propositions are equivalent ($P \iff Q$) if they have the same truth table.

1.3 Logical Operations / Opérations logiques

Definition 2 (Negation / Négation). If P is a proposition, its negation $\overline{P}$ is true when P is false and false when P is true.

Definition 3 (Disjunction / Disjonction). $P \vee Q$ is true if at least one of P or Q is true.

Definition 4 (Conjunction / Conjonction). $P \wedge Q$ is true only if both P and Q are true.

Definition 5 (Implication / Implication). $P \implies Q$ is false only when P is true and Q is false; otherwise true.

Definition 6 (Equivalence / Équivalence). $P \iff Q$ is true if P and Q have the same truth value.

1.4 Basic Logical Laws / Lois logiques de base

1. Commutative Laws / Lois commutatives:

- $P \wedge Q \iff Q \wedge P$
- $P \vee Q \iff Q \vee P$

2. Distributive Laws / Lois distributives:

- $P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) \iff (P \vee Q) \wedge (P \vee R)$

3. Associative Laws / Lois associatives:

- $P \wedge (Q \wedge R) \iff (P \wedge Q) \wedge R$
- $P \vee (Q \vee R) \iff (P \vee Q) \vee R$

4. De Morgan's Laws / Lois de De Morgan:

- $\overline{P \vee Q} \iff \overline{P} \wedge \overline{Q}$
- $\overline{P \wedge Q} \iff \overline{P} \vee \overline{Q}$

5. Contrapositive / Contraposition: $P \implies Q \iff \overline{Q} \implies \overline{P}$

6. Double Negation / Double négation: $\overline{\overline{P}} \iff P$

1.5 Quantifiers / Quantificateurs

Definition 7 (Universal Quantifier / Quantificateur universel). $\forall x, P(x)$ means "for all x , $P(x)$ is true".

Definition 8 (Existential Quantifier / Quantificateur existentiel). $\exists x, P(x)$ means "there exists at least one x such that $P(x)$ is true".

Definition 9 (Uniqueness Quantifier / Quantificateur d'unicité). $\exists! x, P(x)$ means "there exists a unique x such that $P(x)$ is true".

1.5.1 Negation of Quantified Propositions / Négation des propositions quantifiées

1. $\overline{\forall x \in E, P(x)} \iff \exists x \in E, \overline{P(x)}$
2. $\overline{\exists x \in E, P(x)} \iff \forall x \in E, \overline{P(x)}$
3. $\overline{\forall x \in E, \exists y \in E, P(x, y)} \iff \exists x \in E, \forall y \in E, \overline{P(x, y)}$
4. $\overline{\exists x \in E, \forall y \in E, P(x, y)} \iff \forall x \in E, \exists y \in E, \overline{P(x, y)}$

1.6 Proof Techniques / Techniques de preuve

1. **Direct Proof / Preuve directe:** Assume P is true and deduce Q .
2. **Proof by Exhaustion / Preuve par exhaustion:** Prove by verifying all possible cases.
3. **Proof by Contrapositive / Preuve par contraposée:** Prove $P \implies Q$ by proving $\overline{Q} \implies \overline{P}$.
4. **Proof by Contradiction / Preuve par contradiction:** Assume P true and Q false, derive contradiction.
5. **Proof by Counterexample / Preuve par contre-exemple:** Disprove by providing a counterexample.
6. **Proof by Induction / Preuve par induction:**
 - Base case: Prove $P(n_0)$
 - Induction hypothesis: Assume $P(n)$ true
 - Induction step: Show $P(n) \implies P(n+1)$

1.7 Real Number System / Système des nombres réels

1.7.1 Algebraic Properties of $\mathbb{R}$ / Propriétés algébriques de $\mathbb{R}$

- Closure under addition: $a + b \in \mathbb{R}$
- Associative law of addition: $(a + b) + c = a + (b + c)$
- Commutative law of addition: $a + b = b + a$
- Additive identity: $a + 0 = a$
- Additive inverse: $a + (-a) = 0$
- Closure under multiplication: $a \cdot b \in \mathbb{R}$
- Associative law of multiplication: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- Commutative law of multiplication: $a \cdot b = b \cdot a$
- Multiplicative identity: $a \cdot 1 = a$
- Multiplicative inverse: $a \cdot a^{-1} = 1$ ($a \neq 0$)
- Distributive law: $a \cdot (b + c) = a \cdot b + a \cdot c$

1.7.2 Order Structure of $\mathbb{R}$ / Structure d'ordre de $\mathbb{R}$

- Trichotomy property: For $a, b \in \mathbb{R}$, exactly one holds: $a = b$, $a < b$, or $b < a$
- Transitivity: If $a < b$ and $b < c$, then $a < c$
- Addition property: If $a < b$ and $c < d$, then $a + c < b + d$
- Multiplication property: If $a < b$ and $c > 0$, then $ac < bc$
- Reflection property: $a < b \iff -a > -b$

2 Chapter 2: Complex Numbers / Chapitre 2 : Nombres complexes

2.1 Definitions / Définitions

Definition 10. A **complex number** is a number of the form $z = a + bi$, where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$.

- **Real part / Partie réelle:** $\Re(z) = a$
- **Imaginary part / Partie imaginaire:** $\Im(z) = b$
- **Purely imaginary / Purement imaginaire:** $a = 0, b \neq 0$
- **Real case / Cas réel:** $b = 0$
- **Zero / Zéro:** $a = 0, b = 0$

2.2 Algebraic Operations / Opérations algébriques

Let $z_1 = a + bi$, $z_2 = c + di$.

- **Addition / Addition:** $z_1 + z_2 = (a + c) + (b + d)i$
- **Subtraction / Soustraction:** $z_1 - z_2 = (a - c) + (b - d)i$
- **Multiplication / Multiplication:** $z_1 \cdot z_2 = (ac - bd) + (ad + bc)i$

Division / Division (for $z_2 \neq 0$): $\frac{z_1}{z_2} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i$

Equality / Égalité: $z_1 = z_2 \iff a = c \text{ and } b = d$

2.3 Conjugate / Conjugué

Definition 11. The **conjugate** of $z = a + bi$ is $\bar{z} = a - bi$.

Properties / Propriétés:

1. $z \in \mathbb{R} \Rightarrow \bar{z} = z$
2. z purely imaginary $\Rightarrow \bar{z} = -z$
3. $\bar{\bar{z}} = z$
4. $\overline{z + w} = \bar{z} + \bar{w}$
5. $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$
6. $z + \bar{z} = 2\Re(z)$
7. $z - \bar{z} = 2i\Im(z)$

2.4 Modulus (Absolute Value) / Module (Valeur absolue)

Definition 12. $|z| = \sqrt{a^2 + b^2}$

Properties / Propriétés:

1. $|z| = 0 \iff z = 0$
2. $|z| = |\bar{z}| = |-z|$
3. If $\Im(z) = 0$, then $z^2 = |z|^2$
4. $|zw| = |z||w|$, $|\frac{z}{w}| = \frac{|z|}{|w|}$ (for $w \neq 0$)

2.5 Geometric Representation / Représentation géométrique

- **Point / Point:** $z = a + bi$ corresponds to point $M(a, b)$ in complex plane
- **Circle / Cercle:** $|z - z_0| = r$ is circle of center z_0 and radius r
- **Segment / Segment:** $\{z = (1 - t)z_0 + tz_1 \mid t \in [0, 1]\}$
- **Graph / Graphe:** For continuous $f : [a, b] \rightarrow \mathbb{R}$, $G_f = \{z = x + if(x) \mid x \in [a, b]\}$

2.6 Polar Form / Forme polaire

Definition 13. $z = r(\cos \theta + i \sin \theta)$, where:

- $r = |z| = \sqrt{a^2 + b^2}$
- $\tan \theta = \frac{b}{a}$
- **Principal argument / Argument principal:** $\theta \in (-\pi, \pi]$, denoted $\arg(z)$
- **Exponential form / Forme exponentielle:** $z = re^{i\theta}$, where $e^{i\theta} = \cos \theta + i \sin \theta$

Properties of arguments / Propriétés des arguments:

1. $\arg(zz') = \arg(z) + \arg(z') + 2k\pi$
2. $\arg\left(\frac{1}{z}\right) = \arg(\bar{z}) = -\arg(z) + 2k\pi$
3. $\arg\left(\frac{z}{z'}\right) = \arg(z) - \arg(z') + 2k\pi$

2.7 De Moivre's Formula / Formule de De Moivre

Theorem 1. $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$

Theorem 2. For $z = r(\cos \theta + i \sin \theta)$, $z^n = r^n(\cos(n\theta) + i \sin(n\theta))$

2.8 n-th Roots / Racines n-ièmes

Theorem 3. The n -th roots of $z = r(\cos \theta + i \sin \theta)$ are:

$$z^{1/n} = r^{1/n} \left(\cos \left(\frac{\theta + 2k\pi}{n} \right) + i \sin \left(\frac{\theta + 2k\pi}{n} \right) \right), \quad k = 0, 1, \dots, n-1$$

3 Chapter 3: Sequences of Real Numbers / Chapitre 3 : Suites de nombres réels

3.1 Definitions / Définitions

Definition 14. A **sequence** is a function $f : \mathbb{N} \rightarrow \mathbb{R}$, denoted $(U_n)_{n \in \mathbb{N}}$.

Definition 15 (Bounded Sequence / Suite bornée) • **Bounded above / Majorée:** $\exists M, \forall n, U_n \leq M$

- **Bounded below / Minorée:** $\exists m, \forall n, U_n \geq m$

- **Bounded / Bornée:** $\exists k > 0, \forall n, |U_n| \leq k$

Definition 16 (Monotonic Sequence / Suite monotone) • **Increasing / Croissante:** $U_n \leq U_{n+1}$ for all n

- **Strictly increasing / Strictement croissante:** $U_n < U_{n+1}$ for all n

- **Decreasing / Décroissante:** $U_n \geq U_{n+1}$ for all n

- **Strictly decreasing / Strictement décroissante:** $U_n > U_{n+1}$ for all n

- **Constant / Constante:** Increasing and decreasing simultaneously

- **Stationary / Stationnaire:** $\exists n_0, \forall n \geq n_0, U_{n+1} = U_n$

3.2 Convergence / Convergence

Definition 17. (U_n) converges to l if:

$$\forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N}, \forall n \geq N(\epsilon), |U_n - l| < \epsilon$$

Denoted $\lim_{n \rightarrow \infty} U_n = l$.

Theorem 4 (Uniqueness of Limit / Unicité de la limite). If $\lim U_n = l_1$ and $\lim U_n = l_2$, then $l_1 = l_2$.

Theorem 5. Every convergent sequence is bounded.

3.3 Limit Theorems / Théorèmes sur les limites

Theorem 6 (Algebra of Limits / Algèbre des limites). If $\lim U_n = K$ and $\lim V_n = L$, then:

1. $\lim(U_n + V_n) = K + L$
2. $\lim(U_n \cdot V_n) = K \cdot L$
3. If $L \neq 0$, $\lim \frac{U_n}{V_n} = \frac{K}{L}$

Theorem 7 (Squeeze Theorem / Théorème des gendarmes). If $U_n \leq W_n \leq V_n$ for large n and $\lim U_n = \lim V_n = L$, then $\lim W_n = L$.

Theorem 8. If $\lim U_n = 0$ and (V_n) is bounded, then $\lim U_n V_n = 0$.

Theorem 9 (Comparison of Limits / Comparaison des limites). If $U_n \leq V_n$ for large n and both converge, then $\lim U_n \leq \lim V_n$.

3.4 Monotone Sequences / Suites monotones

Theorem 10 (Monotone Convergence Theorem / Théorème de convergence monotone). An increasing sequence bounded above converges to its supremum.

- A decreasing sequence bounded below converges to its infimum.

3.5 Series / Séries

Definition 18. An **infinite series** is $\sum_{n=1}^{\infty} a_n$.

Definition 19. The **partial sum** is $S_n = \sum_{k=1}^n a_k$.

Definition 20. A series **converges** if $\lim S_n = S$ (finite).

Theorem 11 (Geometric Series / Série géométrique). For $a \neq 1$: $\sum_{k=0}^n a^k = \frac{1-a^{n+1}}{1-a}$
For $|a| < 1$: $\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$

3.6 Subsequences / Sous-suites

Definition 21. A **subsequence** of (U_n) is $(U_{\varphi(n)})$ where $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing.

Theorem 12. If (U_n) converges to l , then every subsequence converges to l .

Corollary 1. If two subsequences converge to different limits, (U_n) diverges.

3.7 Adjacent Sequences / Suites adjacentes

Definition 22. (U_n) and (V_n) are **adjacent** if:

1. (U_n) is increasing
2. (V_n) is decreasing
3. $\lim(V_n - U_n) = 0$

Theorem 13. Two adjacent sequences converge to the same limit.

3.8 Cauchy Sequences / Suites de Cauchy

Definition 23. (U_n) is a **Cauchy sequence** if:

$$\forall \epsilon > 0, \exists n_0, \forall p, q \geq n_0, |U_p - U_q| < \epsilon$$

Theorem 14. A sequence converges if and only if it is a Cauchy sequence.

Theorem 15. Every Cauchy sequence is bounded.

3.9 Recurrence Sequences / Suites récurrentes

Definition 24. A recurrence sequence satisfies:

$$\begin{cases} U_0 \text{ given} \\ U_{n+1} = f(U_n), n \geq 0 \end{cases}$$

Definition 25. l is a **fixed point** if $l = f(l)$.

Theorem 16. If f is increasing:

- $U_0 \leq U_1 \Rightarrow (U_n)$ increasing
- $U_0 \geq U_1 \Rightarrow (U_n)$ decreasing

Theorem 17. If f is decreasing, then (U_{2n}) and (U_{2n+1}) are monotone with opposite monotonicity.

Theorem 18. For f decreasing, (U_n) converges iff (U_{2n}) and (U_{2n+1}) are adjacent.

4 Chapter 4: Functions / Chapitre 4 : Fonctions

4.1 Basic Notions / Notions de base

Definition 26. A function $f : A \rightarrow B$ assigns to each $x \in A$ exactly one $y \in B$.

- **Domain / Domaine:** $D(f) \subseteq A$
- **Range / Image:** $H(f) = \{f(x) \mid x \in D(f)\}$
- **Graph / Graphe:** $G_f = \{(x, f(x)) \mid x \in D(f)\}$
- **Equality of functions / Égalité de fonctions:** $f = g$ iff $D(f) = D(g)$ and $f(x) = g(x)$ for all $x \in D(f)$

4.1.1 Bounded Functions / Fonctions bornées

- **Bounded above / Majorée:** $\exists M, \forall x \in D(f), f(x) \leq M$
- **Bounded below / Minorée:** $\exists m, \forall x \in D(f), f(x) \geq m$
- **Bounded / Bornée:** Bounded above and below

4.1.2 Monotonicity / Monotonie

- **Increasing / Croissante:** $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$
- **Strictly increasing / Strictement croissante:** $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$
- **Decreasing / Décroissante:** $x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$
- **Strictly decreasing / Strictement décroissante:** $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$

4.1.3 Composition / Composition

Definition 27. $(g \circ f)(x) = g(f(x))$

Properties / Propriétés:

- **Associativity / Associativité:** $(h \circ g) \circ f = h \circ (g \circ f)$
- **Monotonicity / Monotonie:**
 - f increasing, g increasing $\Rightarrow g \circ f$ increasing
 - f increasing, g decreasing $\Rightarrow g \circ f$ decreasing
 - f decreasing, g increasing $\Rightarrow g \circ f$ decreasing
 - f decreasing, g decreasing $\Rightarrow g \circ f$ increasing

4.1.4 Special Functions / Fonctions spéciales

- **Even function / Fonction paire:** $f(-x) = f(x)$ (symmetric about y -axis)
- **Odd function / Fonction impaire:** $f(-x) = -f(x)$ (symmetric about origin)
- **Periodic function / Fonction périodique:** $\exists p \neq 0, f(x + p) = f(x)$
- **Primitive period / Période primitive:** Smallest positive period

4.2 Limits of Functions / Limites de fonctions

Definition 28 (Limit at a point / Limite en un point). $\lim_{x \rightarrow x_0} f(x) = l$ if:

$$\forall \epsilon > 0, \exists \delta > 0, 0 < |x - x_0| < \delta \Rightarrow |f(x) - l| < \epsilon$$

Definition 29 (One-sided limits / Limites unilatérales) • **Right-hand limit / Limite à droite:** $\lim_{x \rightarrow x_0^+} f(x)$

- **Left-hand limit / Limite à gauche:** $\lim_{x \rightarrow x_0^-} f(x)$

Theorem 19. $\lim_{x \rightarrow x_0} f(x) = l$ iff $\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = l$.

Theorem 20 (Limit Theorems / Théorèmes sur les limites). If $f(x) \geq 0$ near x_0 and $\lim f(x) = l$, then $l \geq 0$.

2. If $f(x) \geq g(x)$ near x_0 and limits exist, then $\lim f \geq \lim g$.
3. (Squeeze Theorem) If $g(x) \leq f(x) \leq h(x)$ near x_0 and $\lim g = \lim h = l$, then $\lim f = l$.
4. If $\lim f = 0$ and g bounded near x_0 , then $\lim fg = 0$.
5. If $\lim_{x \rightarrow x_0} f(x) = l$ and $\lim_{y \rightarrow l} g(y) = m$, then $\lim_{x \rightarrow x_0} g(f(x)) = m$.

4.2.1 Sequential Criterion / Critère séquentiel

Theorem 21. $\lim_{x \rightarrow x_0} f(x) = l$ iff for every sequence (x_n) with $x_n \rightarrow x_0, f(x_n) \rightarrow l$.

4.2.2 Indeterminate Forms / Formes indéterminées

- $\frac{0}{0}, 0 \times \infty, \frac{\infty}{\infty}, +\infty - \infty, \infty^0, 0^0, 1^\infty$

Theorem 22 (Rule for 1^∞ / Règle pour 1^∞). If $f(x) \rightarrow 1, g(x) \rightarrow \pm\infty$, and $\lim(f(x) - 1)g(x) = \lambda$, then $\lim f(x)^{g(x)} = e^\lambda$.

4.3 Continuity / Continuité

Definition 30. f is **continuous at** x_0 if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Definition 31. f is **continuous on an interval** if:

- On $]a, b[$: continuous at every point
- On $[a, b]$: continuous on $]a, b[$, right-continuous at a , left-continuous at b

4.3.1 Discontinuities / Discontinuités

- **First kind / Première espèce:** Left and right limits exist but are not equal or not equal to $f(x_0)$
- **Second kind / Deuxième espèce:** At least one one-sided limit does not exist

Theorem 23 (Properties of Continuous Functions). Sum, difference, product, scalar multiple, absolute value, quotient (if denominator $\neq 0$), and composition of continuous functions are continuous.

4.3.2 Sequential Continuity / Continuité séquentielle

Theorem 24. f continuous at x_0 iff for every sequence $(x_n) \rightarrow x_0$, $f(x_n) \rightarrow f(x_0)$.

4.3.3 Extension by Continuity / Prolongement par continuité

Definition 32. If $\lim_{x \rightarrow x_0} f(x) = l \in \mathbb{R}$ but $x_0 \notin D(f)$, define $\tilde{f}(x_0) = l$ to make f continuous at x_0 .

4.3.4 Intermediate Value Theorem (IVT) / Théorème des valeurs intermédiaires

Theorem 25. If f continuous on $[a, b]$ and $f(a)f(b) < 0$, then $\exists c \in]a, b[$ such that $f(c) = 0$.

Theorem 26 (Generalization / Généralisation). Continuous image of an interval is an interval.

4.3.5 Uniform Continuity / Continuité uniforme

Definition 33. f is **uniformly continuous** on I if:

$$\forall \epsilon > 0, \exists \delta > 0, \forall x, y \in I, |x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$$

Theorem 27 (Heine's Theorem / Théorème de Heine). If f continuous on a closed bounded interval I , then f is uniformly continuous on I .

4.4 Differentiability / Dérivabilité

Definition 34. f is **differentiable** at c if:

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}$$

exists.

Definition 35 (One-sided derivatives / Dérivées unilatérales) • **Left derivative / Dérivée à gauche:** $f'_l(c) = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}$

- **Right derivative / Dérivée à droite:** $f'_r(c) = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}$

Theorem 28. f differentiable at c iff $f'_l(c) = f'_r(c)$ (both exist and are equal).

Theorem 29. Differentiability implies continuity (but not conversely).

4.4.1 Derivative Rules / Règles de dérivation

1. $(f + g)' = f' + g'$

2. $(\alpha f)' = \alpha f'$

3. $(fg)' = f'g + fg'$

4. $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (g \neq 0)$

5. $(g \circ f)' = (g' \circ f) \cdot f'$

6. $(f^n)' = n f^{n-1} f'$

7. **Derivative of inverse / Dérivée de l'inverse:** If f bijective and differentiable at c with $f'(c) \neq 0$, then $(f^{-1})'(f(c)) = \frac{1}{f'(c)}$

4.4.2 Higher-order Derivatives / Dérivées d'ordre supérieur

Definition 36. $f^{(n)}(c)$ is the n -th derivative, defined inductively.

Definition 37 (Class $\mathcal{C}^n$). $f \in \mathcal{C}^n(I, \mathbb{R})$ if:

1. f is n -times differentiable on I
2. $f^{(n)}$ is continuous on I

Theorem 30 (Leibniz Formula / Formule de Leibniz). $(fg)^{(n)} = \sum_{k=0}^n C_n^k f^{(n-k)} g^{(k)}$, where $C_n^k = \frac{n!}{k!(n-k)!}$

4.4.3 Mean Value Theorems / Théorèmes de la moyenne

Theorem 31 (Rolle's Theorem / Théorème de Rolle). If f continuous on $[a, b]$, differentiable on $]a, b[$, and $f(a) = f(b)$, then $\exists c \in]a, b[$ such that $f'(c) = 0$.

Theorem 32 (Mean Value Theorem (MVT) / Théorème des accroissements finis). Under same conditions, $\exists c \in]a, b[$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Theorem 33 (Generalized MVT / Théorème des accroissements finis généralisé). If f, g continuous on $[a, b]$, differentiable on $]a, b[$, and $g'(x) \neq 0$ on $]a, b[$, then $\exists c \in]a, b[$ such that $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$.

4.4.4 L'Hôpital's Rule / Règle de L'Hôpital

Theorem 34. For $\frac{0}{0}$ or $\frac{\infty}{\infty}$ forms: If $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = l \in \mathbb{R} \cup \{\pm\infty\}$, then $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = l$.

4.4.5 Taylor-Lagrange Theorem / Théorème de Taylor-Lagrange

Theorem 35. If f is $\mathcal{C}^n$ on $[a, b]$ and $f^{(n+1)}$ exists on $]a, b[$, then

$$f(b) = \sum_{k=0}^n \frac{(b-a)^k}{k!} f^{(k)}(a) + \frac{(b-a)^{n+1}}{(n+1)!} f^{(n+1)}(c), \quad c \in]a, b[$$